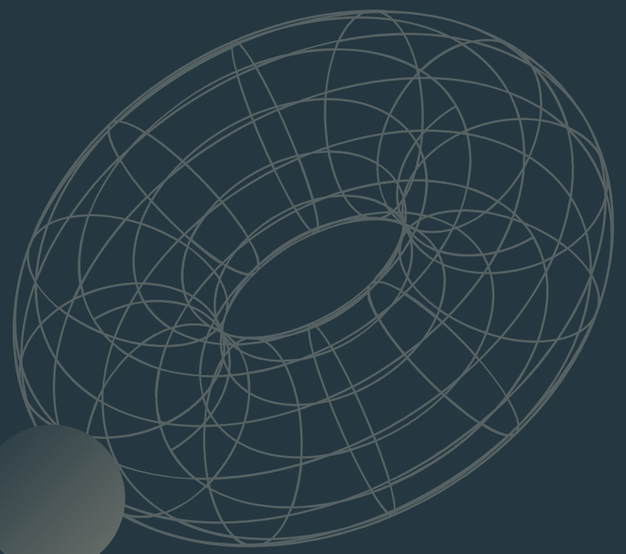
HARDWARE ARCHITECTURE FOR DEEP LEARNING PROJECT

Melanoma Classification for Skin Cancer

Machine Learning Based Skin Cancer Classification

PROBLEM STATEMENT



- Skin cancer is one of the most common cancers worldwide.
- Early detection is crucial as it significantly improves chances of successful treatment.
- Traditional/manual diagnosis relies on dermatologists examining skin lesions, thus can be time-consuming and subjective.
- **Challenges:** limited availability of experts, variation in lesion appearance, risk of misdiagnosis.
- Machine Learning & Deep Learning can assist by providing fast, consistent, and accurate classification.
- **Objective:** Build a model to classify skin lesions into benign (non-cancerous) or malignant (cancerous).

DATASET OVERVIEW

- Source: Melanoma Skin Cancer Dataset (Kaggle).
- Size: ~10,000 dermoscopic images.
- Classes:
  - Benign – non-cancerous lesions.
  - Malignant – cancerous lesions.
- Data Split: Training & Testing sets for model development.
- Visual Diversity: Variations in color, shape, and texture of lesions.

WORKFLOW

The step-by-step pipeline followed for skin cancer classification

Load dataset (Kaggle → Google Colab)

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Data Preprocessing & Augmentation

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Training (with class balancing, callbacks)

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Model building (CNN, ResNet50, VGG16, Ensemble Method)

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Evaluation (accuracy, confusion matrix)

DATA PROCESSING

- **Image Resizing**: All images resized to 224 × 224 for uniformity.
- **Data Augmentation**:
  - Random flip (horizontal/vertical)
  - Rotation
  - Zoom
  - Contrast adjustments
- **Normalization**: Pixel values scaled to range [0,1].
- **Class Balancing**: Used class weights to handle dataset imbalance.

MODEL ARCHITECTURES

Exploring custom-built and pretrained models for classification.

1) Custom CNN

- 3 Convolutional Layers with ReLU activations
- Batch Normalization for stable training
- Dropout layers to prevent overfitting
- Fully Connected (Dense) layer for classification
- Output: 2 classes – Benign, Malignant

2) Transfer Learning – VGG16

- Pretrained VGG16 on ImageNet
- Base layers frozen to retain learned features
- Added custom classification head:
  - Global Average Pooling
  - Dense layer with ReLU + Dropout
  - Output layer (Softmax, 2 classes)

3) Transfer Learning – ResNet50

- Pretrained ResNet50 on ImageNet
- Base layers frozen to retain learned features
- Added custom classification head:
  - Global Average Pooling
  - Dense layers with ReLU
  - Output layer (Softmax, 2 classes)

4) Ensemble Learning

- Combines predictions from multiple models (CNN, VGG16, ResNet50)
- Uses majority voting or weighted averaging
- Improves robustness and overall accuracy
- Output: 2 classes – Benign, Malignant

TRAINING SETUP

- Loss Function: Categorical Cross-entropy
- Optimizer: Adam
- Callbacks Used:
- EarlyStopping → stop training when validation loss stops improving
- ReduceLROnPlateau → lower learning rate when model stagnates
- ModelCheckpoint → save best performing model
- Epochs: 20

RESULTS & PERFORMANCE

1. Custom CNN

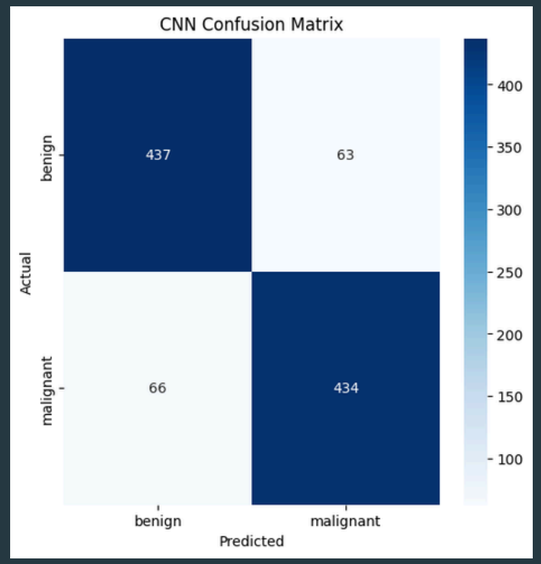
- Accuracy: ~87–89%
- Observations:
  - Strong at detecting benign cases
  - Some misclassifications in malignant cases

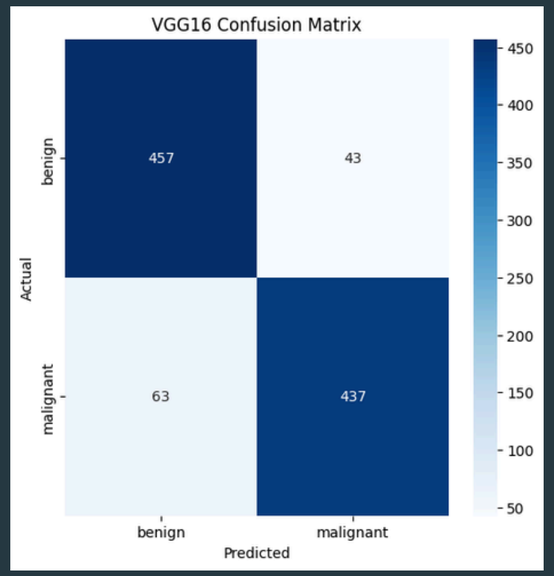
2. VGG16 (Transfer Learning)

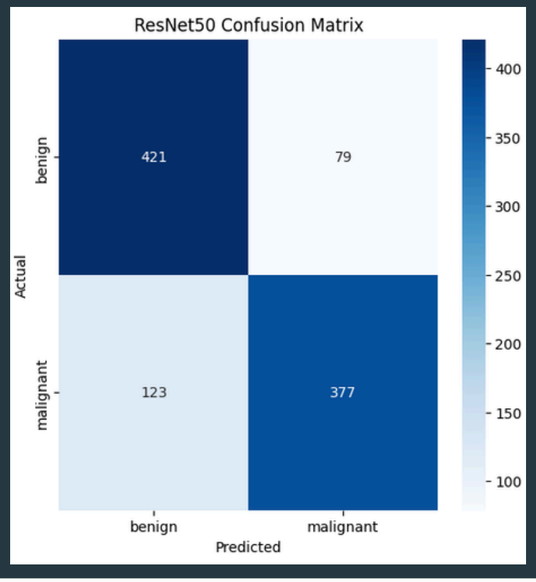
- Accuracy: ~89%
- Observations:
  - Balanced performance overall
  - Slightly lower recall for malignant cases compared to ResNet50

3. ResNet50 (Transfer Learning)

- Accuracy: ~90%
- Observations:
  - Better recall for malignant class
  - More balanced across both classes

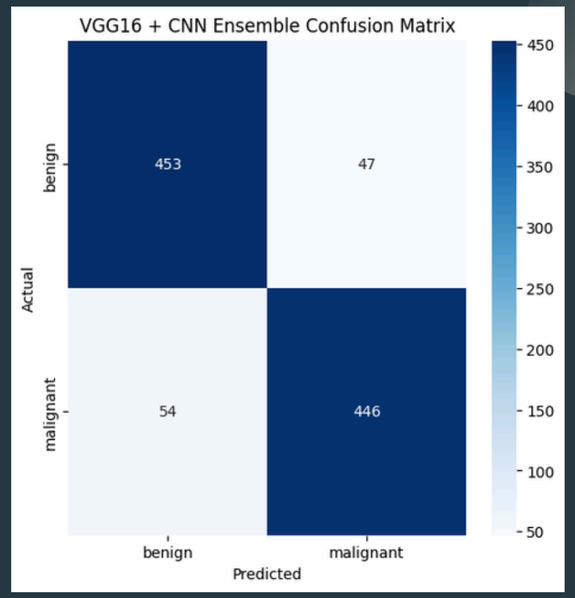


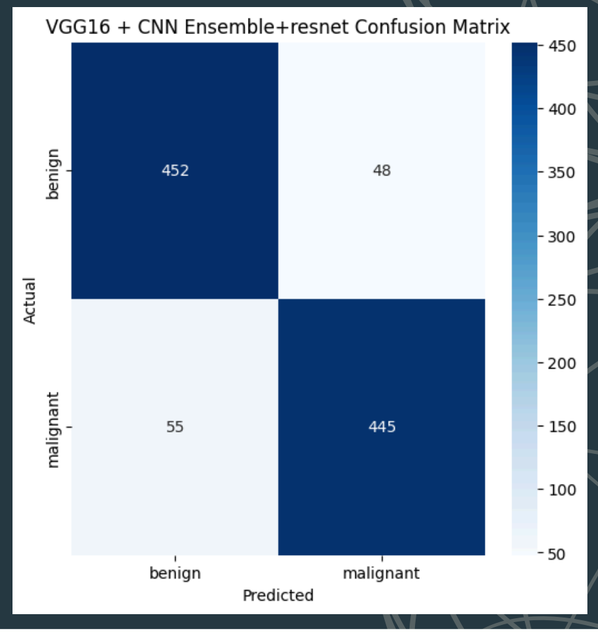




ENSEMBLE MODEL RESULTS

- **VGG16 + CNN Ensemble:**
  - Accuracy: ~90%
  - Precision/Recall balance: ~0.89–0.91
- **VGG16 + CNN + ResNet Ensemble:**
  - Accuracy: ~90%
  - Stable precision/recall across classes
- **Takeaway:**
  - Ensembles provide more reliable predictions.
  - Useful when single models show small biases toward a class.





Conclusion

- Both ResNet50 and Ensemble Learning achieved strong performance.
- Demonstrates the potential of deep learning in medical imaging.
- ResNet50 gave better recall for malignant cases → highly important in cancer detection.

Future Work

- Use larger and more diverse datasets.
- Apply hyperparameter tuning & advanced architectures.
- Explore real-world clinical testing for practical adoption.